2025 秋本科集合论期末（回忆版）

考试时间 120 **分钟 共** 10 **题 总分** 100 **分**

1. (5pts) Write down the Axiom of Pairing using the language of set theory.

2. (5pts) Write down a transfinite recursive definition of ordinal addition and $P(\alpha, \beta)$ for the ordinal multiplication.

3. (5pts) Suppose a sequential tree : $(T, <_T) \subseteq ({}^{<\omega}\omega, \sqsubset)$, where $s \sqsubset t$ if s is a proper initial segment of t. Define a linear ordering that extends $<_T$.

4. (4×5pts) Compute the cardinality of the following sets. No justification is needed. Give your answers in the form of 2^* or $\aleph_*$.

(a) The set of all Borel sets of reals.

(b) The set of all transcendental numbers.

(c) The set of all countable subset of $\aleph_{\omega_1}$, assuming GCH.

(d) The set of all null sets (sets with Lebesgue mearsure zero).

(e) $V_{\omega+9}$, assuming GCH.

5. (4×3pts) Compute the cofinalities of the following ordinals. The additions, multiplications and exponentiations below are all ordinal operations. No justification is needed.

(a) $\aleph_{\omega+\omega^\omega+7}$.

(b) $(\aleph_\omega)^{\omega_1}$.

(c) η, the least fixed point for the ordinal function $f(x) = \omega^{\omega x}$.

6. (3×6pts) Pick out the statements that are NOT consequences of ZFC (including statements that are not true or not provable, from ZFC).

(a) The cantor set is a meager set.

(b) There are more than $\aleph_1$ many reals.

(c) The union of countable many null sets is null.

(d) There are 2^c many Borel meager sets of reals.

(e) There are countable many constant functions from $\mathbb{N}$ to $\mathbb{N}$.

(f) There is a unique linear ordering that is dense, unbounded, complete and satisfies the countable chain condition.

7. (4+5pts) For $\alpha \in Ord$, a function $f : \alpha \to \mathbb{Q}$ is continuous if for all limit ordinal $\beta < \alpha$, $\lim_{i<\beta} f(i)$ exists and $\lim_{i<\beta} f(i) = f(\beta)$.

(a) Present a continuous increasing function $f : \omega + \omega + 1 \to \mathbb{Q} \cap (0, 1]$.

(b) Present a continuous increasing function $f : \omega \times \omega + 1 \to \mathbb{Q} \cap (0, 1]$.

8. (8pts) Show that $cf(\kappa^{cf(\kappa)}) > cf(\kappa)$, for all infinite cardinal κ.

9. (4+6pts)

(a) Write down the definition of a Bernstein set.

(b) Show that every closed subset of a Bernstein set is a null set.

10. (8pts) Show that $\prod_{n<\omega}(n^2 + 1) = 2^{\aleph_0}$.